

PP-complexes and shell structure*

Malhaar Shah & Tonia Bleam

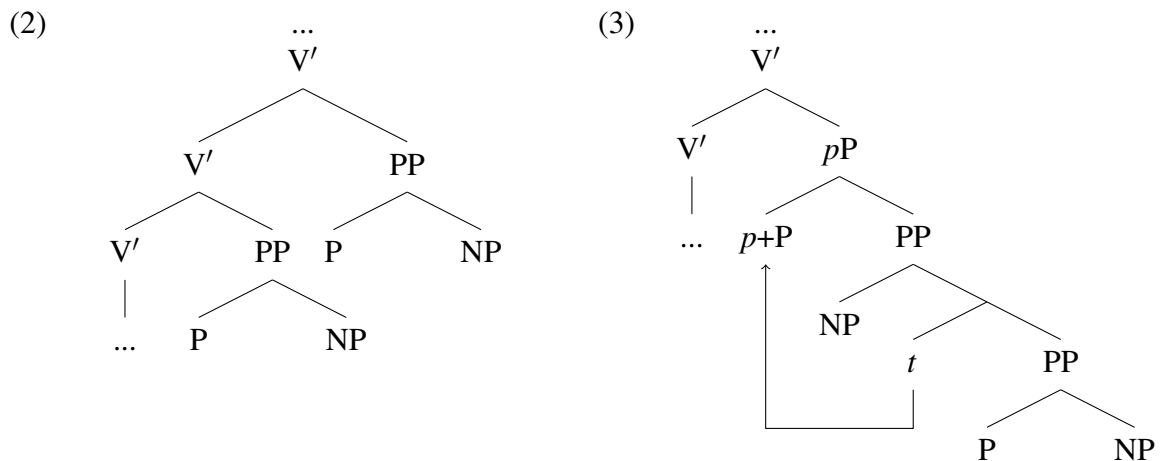
University of Maryland at College Park

1. Introduction

In this paper, we consider cases like (1), where there are, on the surface, two (or more) adjacent PPs.

- (1) a. Jude kicked the ball *past the goalie into the net* in the last minute of extra time.
b. They placed the flagpole *across the mall from the library*.
c. The ferry sailed *across the Channel from Dover to Calais*.

We argue that the grammar allows two distinct derivations for two adjacent PPs. We call the derivation in (2) ‘PP-stacking’, and the derivation in (3) a ‘PP-complex’.¹



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¹‘PP-stacking’, for us, refers to any derivation where both PPs are adjoined to VP. We do not commit to the possibility of right-adjunction, but use it for expository convenience.

Examples we discuss below show that some sentences, like (1a) are ambiguous between a PP-stacking and a PP-complex derivation, but other sentences, like (1b), allow *only* a PP-complex derivation. This is supported by arguments from entailment, constituency, and binding. This has as a notable consequence that there can be a one-to-many relationship between meanings and syntactic derivations. In addition, we demonstrate that the facts are compatible with a binary-branching model of grammar, in contrast to the conclusions from the previous work which has drawn attention to the phenomena discussed below.

This paper is organized as follows. In §2, we provide an overview of two important previous contributions to this investigation, from Jackendoff (1973) and Dowty (1979). These authors argue, on the basis of syntactic and semantic considerations, that there are non-PP-stacking parses of multiple PP-modification. While we find their conclusion compelling, we raise empirical and conceptual issues with their proposed analyses in §3. One of these issues is that Jackendoff's analysis requires negotiating on the assumption that phrase-structure is (at most) binary-branching. In §4, we motivate and provide an alternative analysis that maintains the assumption of binary-branching, by adding shell-structure above PP, inspired by Larson (1988). We comment here on the relation between syntax and semantics, and point out some predictions of our analysis before concluding in §5, which also notes some open issues surrounding semantics, cartography, and cross-linguistic variation.

2. Background: Jackendoff & Dowty's arguments against PP-stacking

In this section, we lay out some background. These facts and arguments are presented by Jackendoff (1973) and Dowty (1979) as evidence against a PP-stacking parse of the relevant sentences. We recount these arguments here, before discussing some issues with the proposals that Jackendoff and Dowty make in the next section. The facts that Jackendoff raises constitute a large part of the empirical focus of this paper.

2.1 Jackendoff's arguments from constituency

Jackendoff (1973) provides arguments from clefting, locative inversion, and adverb placement. That two PPs *can* function as a constituent is suggested by their serving as the pivot of a cleft (4a) and by their inverting as a unit (4b) (examples adapted).

- (4) They placed the flag across the mall from the library.
- a. It was *across the mall from the library* that they placed the flag. (clefting)
 - b. *Across the mall from the library* stood a flag. (loc. inv.)

Furthermore, the examples we have provided allow no derivation where the PPs do not form a constituent. When only one of the two PPs is inverted/clefted, the results are ungrammatical:²

²In Jackendoff's analysis, the ungrammaticality of (6) follows from *from the library's* not being a constituent. In our analysis, to be discussed below, *from the library* does form a constituent. We think that it is

PP-Complexes and shell structure

- (5) a. *It was *across the mall* that they placed the flag from the library.
b. **Across the mall* stood a flag from the library.
- (6) a. *It was *from the library* that they placed the flag across the mall.
b. **From the library* stood a flag across the mall.

Jackendoff also observes that an adverb should be able to adjoin between the PPs, if they are independent verbal adjuncts (other things equal; cf. Cinque 1999). But this is not borne out. An adverb can appear on either side of the PPs, but not in between:

- (7) The flag stands
(imposingly) across the mall (*imposingly) from the library (imposingly).

These syntactic facts are straightforwardly explained if *across the mall* does not form a PP-constituent to the exclusion of *from the library*.

2.2 Dowty's arguments from entailment

Dowty (1979), working in a Montague Grammar framework, notices a different issue. He discusses a problem that a PP-stacking parse would cause for his independently-motivated decompositions for *from* and *to*.³ Dowty takes transitive V modification by goal PPs like *to the fence* (8a) and source PPs like *from the fence* (8b) to entail causation (p. 208ff):

- (8) a. John moved a rock to the fence.
 ≈ The rock to be at the fence as a result of [John moving the rock].
b. John moved a rock from the fence.
 ≈ The rock came to not be at the fence as a result of [John moving the rock].

But extending this sort of logic to sentences like (9) turns out to be problematic if (9) is assigned a PP-stacking parse.

- (9) John drove a car from Boston to Chicago.

Under a PP-stacking parse, *to Chicago* takes scope over *drove a car from Boston*, yielding implausibly strong truth-conditions (given Dowty's assumptions):

not constituency, but a constraint on movement which blocks (6): subextraction of a PP from a PP would be non-minimal.

³It should be remembered that, for Dowty, the decomposition is purely in the metalanguage used to describe the language under consideration (English), and thus serves no purpose other than to make perspicuous the entailments associated with a lexical item (and is not intended as a claim about the meaning of the English word, or about the psychology of English speakers).

- (10) John [[drove a car from Boston] to Chicago]

Predicted meaning: The car came to be in Chicago as a result of [the car coming to not be in Boston as a result of [John's driving the car]]

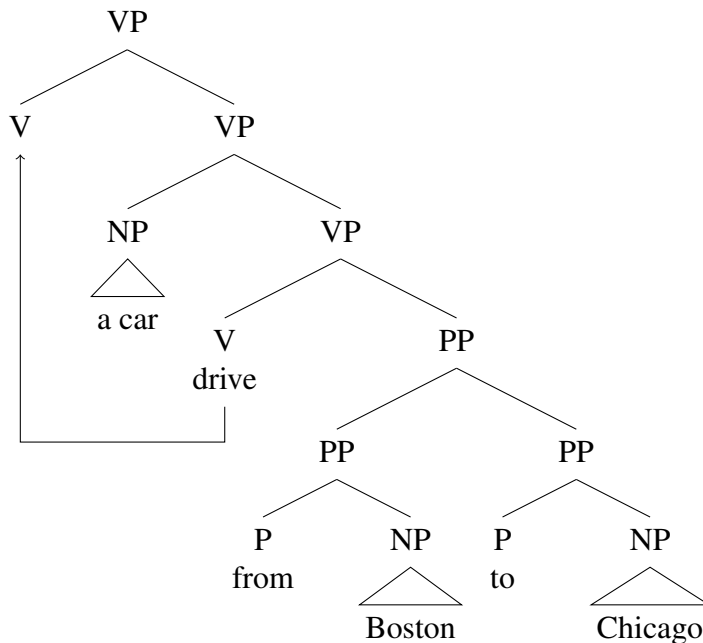
Dowty points out that (9)'s truth-conditions are weaker: there need be no causal connection between the point of departure and the destination. It need only be the case that John's driving the car leads the car to be in Chicago, and that the driving originated in Boston. The problematic reading arises (under Dowty's assumptions) because *to Chicago* takes scope over the VP modified by *from Boston*. Dowty thus rejects the PP-stacking parse. He also notes in passing (p. 216) the concomitant syntactic facts, i.e., that *from Boston to Chicago* is a constituent according to the clefting test (as predicted by his semantic analysis).⁴

Jackendoff and Dowty ultimately propose different syntactic analyses for these expressions. However, both lead to conceptual and empirical problems. In the next section, we outline the proposals and the problems they face, starting with Dowty's before turning to Jackendoff's.

3. Problems for the previous proposals

Under Dowty's proposal, the prepositions combine with their individual arguments before combining with one another, and then with a transitive verb:

- (11)
- (to be rejected)



If the PPs conjoin, neither PP takes scope over the other. The problematic reading discussed above then does not arise, allowing Dowty to maintain his decompositions of *from* and *to*.

⁴Switching the syntactic heights of the PP-constituents does not affect the argument for Dowty, as the causation entailed by either PP must be prevented from take scope over the other.

It is worth noting that one might have come to a similar hypothesis on independent, syntactic grounds. Kayne (1994:22ff) discusses the prohibition on phrasal multiple-adjunction imposed by the Linear Correspondence Axiom (LCA). The LCA forbids anything like the (2) parse, with two VP adjuncts, but it permits a derivation where one PP is adjoined to another PP, which is itself adjoined to VP (Kayne himself schematically notes this kind of derivation).⁵

Even if one grants that the internal argument's merger with V may be delayed in this way (just as in Larson 1988, who takes adjuncts to be sisters to V), there is a problem *within* the phrase-structure of the PP. Under Dowty's proposal, neither NP c-commands the other. But there is evidence that one of the NPs does, in fact, asymmetrically c-command the other.

That there is asymmetric c-command between the NPs is shown, for instance, in variable binding being available from left-to-right (12a) but not from right-to-left (12b).

- (12) a. The worst vendor stands across every intersection₁ from its₁ busiest corner.
(Tessa Culleton, p.c.)
b. * They installed a streetlight across its₁ driveway from every house₁.

Dowty's structure wrongly predicts that neither NP takes scope over the other, or that either may. Thus some syntactic asymmetry must be introduced (assuming that linear order *per se* may not be invoked in establishing the possibilities for binding; Reinhart 1983).

Jackendoff's hypothesis, on the other hand, is able to cope with the Binding facts. It posits an asymmetry between the NPs. He proposes that these cases are ternary-branching:

- (13) [PP across_P [NP every intersection] [PP from its busiest corner]]

This structure has the right kind of asymmetry between the NPs to capture the binding facts: *every intersection* asymmetrically c-commands *its busiest corner*.⁶ This is an advantage over Dowty's hypothesis. But it requires us to relax an otherwise attractive restriction on phrase structure: that it be at most binary-branching (Kayne 1981, 1994, Larson 1988, Chomsky 1995; see also Pesetsky 1995).

The binary-branching view is attractive for at least two reasons. It severely restricts the class of possible derivations, and it brings the structure of the syntactic derivation closer

⁵Again, we have put aside issues relating to right-adjunction to VP.

⁶In conversation with Matthew Hewett, it was noted that there is also an asymmetry in extractability:

- (i) a. ? Which city does Calais sit across the Channel from?
b. * Which sea does Calais sit across from Dover?

While we think this fact neither supports nor challenges our central claim, the PP-complex constituent structure we propose suggests an analysis of this in terms of a Left Branch Effect. We do not investigate this further here.

to mirroring the semantic composition (a point emphasized by Larson 2014).⁷ For theories that assume structure is generated by Merge, as defined in Chomsky (1995:Ch. 4), the binary-branching restriction is derived in an even stronger form, as even unary projections are disallowed.

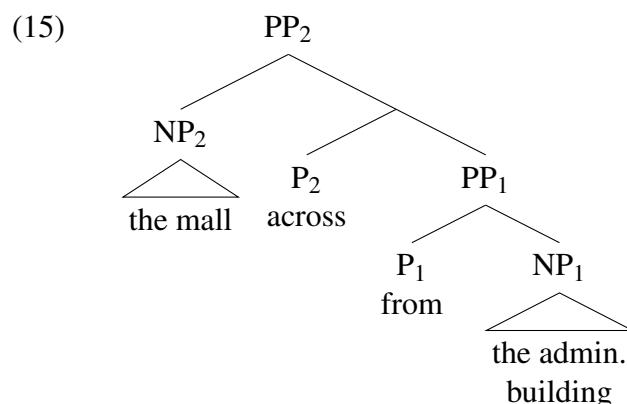
We now turn to our proposal, which is intended to maintain the strengths of Jackendoff's proposal (in that it posits an asymmetry between the NPs within the PP), without relaxing the binary-branching constraint.

4. The PP-shell hypothesis

We begin by noting a contrast between two uses of *across-NP-from-NP*:

- (14) a. The ferry sailed across the Channel from Dover.
 $\models$ The ferry sailed across the Channel.
 $\models$ The ferry sailed from Dover.
- b. McKeldin Library stands across the mall from the Administration Building.
 $\not\models$ McKeldin Library stands across the mall.
 $\not\models$ McKeldin Library stands from the mall.

The first use – the ‘Path’ use (a) – is compositional. We assume that in the ‘Place’ use (b), that *across* selects *from* (perhaps reflecting the fact that, as Nam 1995:107 puts it, *across-NP-from-NP* cannot be defined in terms of *across* or *from*). We therefore assume that the base-structure for the Place use must be as in (15):

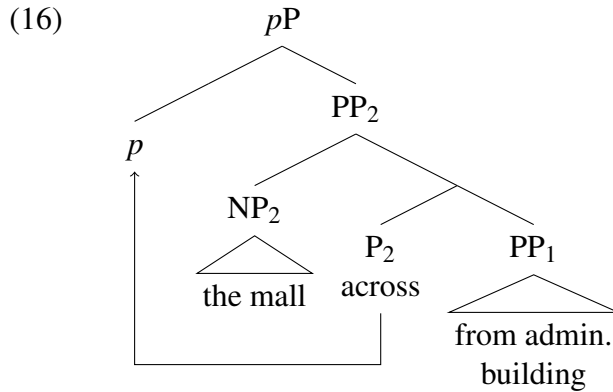


However, the structure in (15) is inadequate. It captures the selection relation locally, it leads to the wrong word-order. To remedy this, we assume a head *p*, higher than *PP*₂, in the spirit of van Riemsdijk (1990). We assume further that *p* triggers head movement⁸ of

⁷As Larson (1988:fn. 47) notes, the binary-branching assumption in syntax is closely related to the semantic assumption in Montague (1973), that all functions are Curried.

⁸The head-movement is required only for a change in word-order. We do not rule out other implementations of this effect (e.g., linearization features in a Spanning framework, as outlined in Svenonius 2016).

its complement (for similar ideas, see Matsubara 2000, Kayne 2005 on P leapfrogging over its Spec, and Merchant 2009:151).⁹



Thus, we capture the constituency facts raised in Section 2.1 by treating expressions like *across the mall from the library* as a *pP* constituent. Furthermore, we do this without relaxing the binary-branching constraint on phrase structure.

In this case, because of the selectional properties involved between *across* and *from*, a local relationship is forced. Only a PP-complex parse may be assigned in the Place use of *across the mall from the library*, and a PP-stacking parse is disallowed. For other pairs of prepositions (e.g., *past/into*), which are not in a selectional relationship, there is nothing to force the PP-complex derivation. It seems that similar facts hold for *from-until*.

When there is nothing like selection to *force* the PP-complex parse, it seems that multiple parses are available – both PP-stacking *and* PP-complexes. This is shown by clefting:

- (17) Jude kicked the ball *past the goalie into the net* in the last minute of extra time.
- It was past the goalie that Jude kicked the ball into the net.
 - It was into the net that Jude kicked the ball past the goalie.
 - It was past the goalie into the net that Jude kicked the ball.

While (c) shows that a PP-complex parse is available, (a) and (b) show that it is not the only available parse. Furthermore, in these cases, adverbs *can* intervene between the PPs:

- (18) Jude kicked the ball (forcefully) past the goalie (forcefully) into the net (forcefully) in the last minute of extra time.

Thus we submit that these cases are syntactically ambiguous between PP-stacking parses and PP-complex parses.

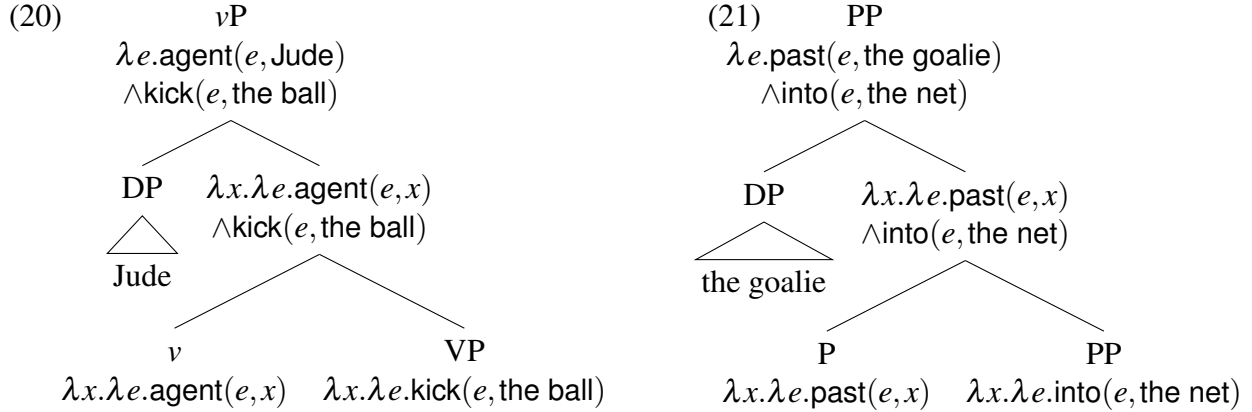
⁹One may ask whether the existence of *p* can be independently motivated. We speculate that Case theory provides a rationale. If Case is assigned under c-command by a licensor (potentially mediated by a relation Agree; Chomsky 2000), *across* is unable to Case-license the NP merged in its Specifier position. *p* must be merged as ‘rescue’, to assign Case to NP₂.

This has the theoretically notable consequence that there can be a one-to-many relationship between meanings and syntactic derivations. The meaning in (19)¹⁰ can be expressed by means of both a PP-complex or a PP-stacking derivation, as shown in (17).

(19) $\exists e. \text{kick}(e) \wedge \text{agent}(e, \text{Jude}) \wedge \text{theme}(e, \text{the ball}) \wedge \text{past}(e, \text{the goalie}) \wedge \text{into}(e, \text{the net})$

This is similar to (though not identical to) a point that has been emphasized in the Categorical Grammar tradition, in which it is routine to say that there are many (sometimes even infinitely many) parses of a string that all yield the same meaning (see e.g. Jacobson 2014:§11.3; cf. Steedman 1991).

The PP-stacking parse can yield the meaning in (19) if there is a rule of composition which allows the conjunction of event predicates (or if PPs can be type $\langle\langle v, t \rangle, \langle v, t \rangle\rangle$). On the other hand, the PP-complex parse can also yield (19) if there is a rule of composition which allows thematic roles (type $\langle e, \langle v, t \rangle \rangle$) to compose with an event-predicate before the individual that saturates the role. One such rule is Kratzer's (1996) Event Identification rule, designed for thematic role assignment to a specifier as in (20). This same rule allows the composition shown in (21), still yielding the meaning in (19).



We now explore a prediction of our proposal before concluding. Prepositions are, in many cases, l-selected. Merchant (2019) provides many cases where lexical items like *proud/pride* idiosyncratically (i.e., l(exically)) select particular prepositions:

- (22)
- a. I am *proud* {*of*/**in*/**on*} my work.
 - b. I take *pride* {*in*/**of*/**on*} my work.
 - c. I *pride* myself {*on*/**of*/**in*} my work.

Our hypothesis predicts that there could be no head which l-selects a PP-complex like *across-NP-from-NP*. Just as *proud* l-selects *of*, a head could l-select *across*, and *across* could in turn, l-select *from*. But to select a PP-complex, a head would have to l-select *p*,

¹⁰Semantic values are indicated in sans serif.

which would leave it unable to further select *across*. Thus, we predict that PP-complexes are never l-selected.¹¹

5. Conclusion

This paper investigated the syntactic properties (and more briefly, the semantic properties) of phrases like *across the mall from the library*. We argued, following earlier work by Jackendoff (1973) and Dowty (1979), that these phrases necessarily form constituents. We argued that the fact about *across*-NP-*from*-NP that *forces* it to be a constituent is that *across* selects *from*. We analyzed PPs as allowing an extra head *p* above them, which allows a preposition to head-move and linearize to the left of its Specifier argument. This is much as in Larson's (1988) treatment of verbal 'shells'. We are able to capture the facts discussed above in a binary-branching model of syntax, unlike previous proposals.

We argued further that expressions like *past the goalie into the net* are ambiguous between a parse where they do not form a constituent (2) and one where they do (3). This had the consequence that there is sometimes a one-to-many relation between meanings and syntactic derivations.

However, we have left open many issues, like whether *p* contributes to semantics (see Kratzer 1996 on Voice, but see Larson 2014:76-80 for a critical discussion of a semantics for *v*). The spuriously ambiguous cases in (17) suggest that *p* does not always contribute to semantics. We have also left open the connection to work which has found evidence for several functional projections within the PP (see for instance, the essays collected in Cinque and Rizzi 2010). Lastly, we have not addressed what role (if any) *p* plays in handling the difference between postpositional and prepositional languages. We leave to future work the question of whether there is evidence for Ps like *across* which take both a PP argument *from the library* and NP argument *the mall* in these languages (or other prepositional languages, for that matter).

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¹¹This is interesting because, descriptively, a head *is* able to l-select two prepositions. Sebastián Mancha (p.c.) points out the potential relevance of cases like (i):

- (i) a. I looked *down on* him.
- b. I looked *up to* him.
- c. I watched *out for* the package for my neighbor.
- d. I stocked *up on* groceries before the snowstorm.
- e. I put *up with* the change in scheduling.

It seems that the grammar has a mechanism which allows a head to regulate which two prepositions it occurs with. Yet, we are not aware of any analogous cases where a head selects not P-P-NP but P-NP-P-NP. It seems that this mechanism does not extend to two prepositional *phrases*, where the grammar ensures that a head occurs with particular Ps in P-NP-P-NP. This follows if *p* must be involved in P-NP-P-NP constituents.

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Malhaar Shah & Tonia Bleam

mpshah@umd.edu, tbleam@umd.edu